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Generative - AI

Experiment No. 1

Product Description for a Smart Fitness Watch

Aim

To design Zero-shot, One-shot, and Few-shot prompts to generate a product description for a Smart Fitness Watch using a Large Language Model (LLM), and to observe how the quality and style of the generated description change with each prompting technique.

Algorithm

1. Start the program and import the required library for accessing the LLM API.
2. Initialize the API client with the model name and authentication key.
3. Define a function `generate_response()` that sends a given prompt to the LLM and returns the generated text.
4. Construct a Zero-shot prompt that directly instructs the model to write a product description without providing any example.
5. Construct a One-shot prompt that supplies one sample product description as a reference before asking for the new description.
6. Construct a Few-shot prompt that supplies two or more sample product descriptions as reference before asking for the new description.
7. Pass each of the three prompts to `generate_response()` and store the results.
8. Display the Zero-shot, One-shot, and Few-shot outputs.
9. Compare the three outputs for creativity, structure, and relevance to the product.
10. Stop.

Python Program

```
import anthropic
```

```
client = anthropic.Anthropic(api_key="YOUR_API_KEY")
```

```
def generate_response(prompt, max_tokens=200):  
    response = client.messages.create(  
        model="claude-3-5-sonnet-20241022",  
        max_tokens=max_tokens,  
        messages=[{"role": "user", "content": prompt}]  
    )  
    return response.content[0].text.strip()
```

```
# ----- Zero-shot Prompt -----
```

```
zero_shot_prompt = """Write a product description for a Smart Fitness Watch."""
```

```
# ----- One-shot Prompt -----
```

```
one_shot_prompt = """Example:
```

```
Product: Wireless Bluetooth Earbuds
```

```
Description: Experience true wireless freedom with crystal-clear  
sound, 24-hour battery life, and a sweat-resistant design perfect  
for workouts and daily use.
```

```
Now write a similar product description for a Smart Fitness Watch."""
```

```
# ----- Few-shot Prompt -----
```

```
few_shot_prompt = """Example 1:
```

```
Product: Wireless Bluetooth Earbuds
```

```
Description: Experience true wireless freedom with crystal-clear  
sound, 24-hour battery life, and a sweat-resistant design perfect  
for workouts and daily use.
```

```
Example 2:
```

```
Product: Portable Power Bank
```

```
Description: Stay charged on the go with our 20000mAh Portable  
Power Bank, offering fast charging, multiple ports, and a compact  
design that fits easily into any bag.
```

```
Now write a similar product description for a Smart Fitness Watch."""
```

```
print("ZERO-SHOT OUTPUT:\n", generate_response(zero_shot_prompt))  
print("\nONE-SHOT OUTPUT:\n", generate_response(one_shot_prompt))  
print("\nFEW-SHOT OUTPUT:\n", generate_response(few_shot_prompt))
```

Input

Product Name : Smart Fitness Watch

Prompting Techniques Used : Zero-shot, One-shot, Few-shot

(Prompts as defined in the Python program above)

Output

ZERO-SHOT OUTPUT:

Stay on top of your health goals with the Smart Fitness Watch. Track your heart rate, sleep, and daily steps in real time, monitor workouts across 20+ sport modes, and get instant call and message alerts on your wrist. With a 7-day battery life and a sleek, water-resistant design, it keeps you connected and motivated wherever you go.

ONE-SHOT OUTPUT:

Take control of your fitness journey with our Smart Fitness Watch, featuring continuous heart-rate and SpO2 monitoring, built-in GPS, and a vibrant AMOLED display. With up to 10 days of battery life and a rugged, sweat-resistant build, it is the perfect companion for runners, gym-goers, and everyday adventurers.

FEW-SHOT OUTPUT:

Meet your new fitness companion - the Smart Fitness Watch. Track steps, calories, heart rate, and sleep patterns with precision, receive smart notifications on the go, and choose from 15+ workout modes. Built with a durable, water-resistant design and a long-lasting battery, it keeps you moving and motivated all day, everyday.

Observation: The Zero-shot response is generic, the One-shot response adopts the tone and structure of the given example, and the Few-shot response is the most consistent and marketing-ready, closely matching the style of both reference examples.

Experiment No. 2

Blog on "Applications of Artificial Intelligence in Healthcare"

Aim

To design Zero-shot, One-shot, and Few-shot prompts to generate a 200-word blog on "Applications of Artificial Intelligence in Healthcare" using an LLM, and to compare the outputs generated by each technique.

Algorithm

1. Start the program and import the required library for accessing the LLM API.
2. Initialize the API client with the model name and authentication key.
3. Define a function `generate_response()` that sends a prompt to the LLM and returns the generated text.
4. Construct a Zero-shot prompt asking the model to directly write a 200-word blog on the given topic.
5. Construct a One-shot prompt that provides a short sample blog paragraph on a related topic before requesting the new blog.
6. Construct a Few-shot prompt that provides two sample blog paragraphs on related topics before requesting the new blog.
7. Send each prompt to `generate_response()` and store the generated blogs.
8. Display the Zero-shot, One-shot, and Few-shot blog outputs.
9. Compare the outputs for word count accuracy, structure, and depth of content.
10. Stop.

Python Program

```
import anthropic

client = anthropic.Anthropic(api_key="YOUR_API_KEY")

def generate_response(prompt, max_tokens=400):
    response = client.messages.create(
        model="claude-3-5-sonnet-20241022",
        max_tokens=max_tokens,
        messages=[{"role": "user", "content": prompt}]
    )
    return response.content[0].text.strip()

topic = "Applications of Artificial Intelligence in Healthcare"

# ----- Zero-shot Prompt -----
zero_shot_prompt = f"Write a 200-word blog on '{topic}'."

# ----- One-shot Prompt -----
```

```
one_shot_prompt = f"""Example blog opening (topic: AI in Education):
"Artificial Intelligence is transforming classrooms by personalizing
learning paths, automating grading, and providing intelligent tutoring
systems that adapt to each student's pace."
```

```
Now write a complete 200-word blog on '{topic}' in a similar
informative and engaging style."""
```

```
# ----- Few-shot Prompt -----
```

```
few_shot_prompt = f"""Example 1 (topic: AI in Education):
"Artificial Intelligence is transforming classrooms by personalizing
learning paths, automating grading, and providing intelligent tutoring
systems that adapt to each student's pace."
```

```
Example 2 (topic: AI in Finance):
```

```
"Artificial Intelligence is reshaping finance through fraud detection,
algorithmic trading, and robo-advisors that make investment guidance
accessible to everyone."
```

```
Now write a complete 200-word blog on '{topic}' following the same
informative, engaging tone used above."""
```

```
print("ZERO-SHOT OUTPUT:\n", generate_response(zero_shot_prompt))
print("\nONE-SHOT OUTPUT:\n", generate_response(one_shot_prompt))
print("\nFEW-SHOT OUTPUT:\n", generate_response(few_shot_prompt))
```

Input

Topic : Applications of Artificial Intelligence in Healthcare

Required Length : 200 words

Prompting Techniques Used : Zero-shot, One-shot, Few-shot

Output

ZERO-SHOT OUTPUT :

Artificial Intelligence (AI) is revolutionizing the healthcare industry by improving diagnosis, treatment, and patient care. AI- powered imaging tools can detect diseases such as cancer and diabetic retinopathy from scans with high accuracy, often catching abnormalities earlier than the human eye. Machine learning models analyze patient records to predict disease risk, enabling preventive care rather than reactive treatment. In hospitals, AI-driven chatbots and virtual assistants help patients book appointments, answer basic health queries, and monitor symptoms remotely. Robotic surgery systems, guided by AI algorithms, allow surgeons to perform minimally invasive procedures with greater precision. Drug discovery, once a decade-long process, is being accelerated through AI models that simulate molecular interactions and identify

promising compounds faster. Wearable devices equipped with AI continuously track vital signs like heart rate and oxygen levels, alerting users and doctors to potential health issues in real time. AI also plays a key role in hospital administration, optimizing staff schedules, managing electronic health records, and reducing operational costs. Despite challenges around data privacy and regulatory approval, the integration of AI in healthcare continues to expand, promising faster diagnoses, personalized treatment plans, and improved outcomes for patients worldwide, while reducing the burden on healthcare professionals.

ONE-SHOT OUTPUT

Artificial Intelligence is transforming healthcare by bringing speed, precision, and personalization to patient care. Much like AI adapts to a student's pace in education, AI systems in healthcare adapt to a patient's unique medical history to recommend personalized treatment plans...

FEW-SHOT OUTPUT

Just as Artificial Intelligence has reshaped education and finance through personalization and automation, it is now redefining healthcare. From AI-assisted diagnostics that flag early signs of disease to robo-guided surgical tools that improve precision, the technology touches nearly every corner of modern medicine...

Observation: The Zero-shot output is well-structured but generic in tone; the One-shot output mirrors the style of the given example opening; the Few-shot output shows the most consistent tone and transition style, drawing parallels across domains as shown in the two examples.

Experiment No. 3

Article Summarization in 50 Words

Aim

To design Zero-shot, One-shot, and Few-shot prompts to summarize a given article into 50 words, and to compare the generated summaries based on accuracy, completeness, and readability.

Algorithm

1. Start the program and import the required library for accessing the LLM API.
2. Initialize the API client with the model name and authentication key.
3. Store the input article text in a variable named article.
4. Define a function generate_response() that sends a prompt to the LLM and returns the generated text.
5. Construct a Zero-shot prompt that asks the model to summarize the article in 50 words without any example.
6. Construct a One-shot prompt that shows one example of an article and its 50-word summary before asking the model to summarize the given article.
7. Construct a Few-shot prompt that shows two examples of articles with their 50-word summaries before asking the model to summarize the given article.
8. Send each prompt to generate_response() and store the three summaries.
9. Display the Zero-shot, One-shot, and Few-shot summaries.
10. Compare the three summaries manually or using a scoring rubric based on accuracy, completeness, and readability, and tabulate the result.
11. Stop.

Python Program

```
import anthropic
```

```
client = anthropic.Anthropic(api_key="YOUR_API_KEY")
```

```
def generate_response(prompt, max_tokens=120):  
    response = client.messages.create(  
        model="claude-3-5-sonnet-20241022",  
        max_tokens=max_tokens,  
        messages=[{"role": "user", "content": prompt}]  
    )  
    return response.content[0].text.strip()
```

```
article = """
```

Artificial Intelligence (AI) is rapidly changing the way industries operate across the globe. From healthcare to finance, education to transportation, AI-powered systems are automating repetitive

tasks, uncovering patterns in massive datasets, and enabling faster, data-driven decisions. In healthcare, AI assists doctors in diagnosing diseases early. In finance, it detects fraudulent transactions in real time. In transportation, self-driving vehicles rely on AI to navigate roads safely. However, the growth of AI also raises concerns about job displacement, data privacy, and the need for clear ethical guidelines. Governments and organizations worldwide are now working on policies to ensure AI is developed and used responsibly, balancing innovation with accountability so that its benefits are shared widely while its risks are minimized.

""

----- Zero-shot Prompt -----

zero_shot_prompt = f"Summarize the following article in exactly 50 words:\n{article}"

----- One-shot Prompt -----

one_shot_prompt = f""Example:

Article: "Renewable energy sources such as solar and wind are becoming cheaper and more efficient, helping countries reduce dependence on fossil fuels and cut carbon emissions."

Summary (50 words): Renewable energy, especially solar and wind, is becoming more affordable and efficient. This shift helps nations lower fossil fuel dependence and reduce carbon emissions, supporting global efforts to combat climate change while promoting sustainable, long-term economic growth through cleaner power generation methods.

Now summarize the following article in exactly 50 words:

{article}""

----- Few-shot Prompt -----

few_shot_prompt = f""Example 1:

Article: "Renewable energy sources such as solar and wind are becoming cheaper and more efficient, helping countries reduce dependence on fossil fuels and cut carbon emissions."

Summary (50 words): Renewable energy, especially solar and wind, is becoming more affordable and efficient. This shift helps nations lower fossil fuel dependence and reduce carbon emissions, supporting global efforts to combat climate change while promoting sustainable, long-term economic growth through cleaner power generation methods.

Example 2:

Article: "E-commerce has grown rapidly worldwide, with more consumers shopping online for convenience, better prices, and wider product selection than traditional retail stores can offer."

Summary (50 words): E-commerce has expanded rapidly as consumers increasingly prefer online shopping for its convenience, competitive pricing, and broader product variety. This shift challenges traditional retail stores to adapt, prompting many businesses to strengthen their digital presence and improve online customer experience to remain competitive.

Now summarize the following article in exactly 50 words:

{article}""

print("ZERO-SHOT SUMMARY:\n", generate_response(zero_shot_prompt))


```
print("\nONE-SHOT SUMMARY:\n", generate_response(one_shot_prompt))
print("\nFEW-SHOT SUMMARY:\n", generate_response(few_shot_prompt))
```

Input

Article : "Artificial Intelligence (AI) is rapidly changing the way industries operate across the globe. From healthcare to finance, education to transportation, AI-powered systems are automating repetitive tasks, uncovering patterns in massive datasets, and enabling faster, data-driven decisions... Governments and organizations worldwide are now working on policies to ensure AI is developed and used responsibly."

Required Summary Length : 50 words

Prompting Techniques Used : Zero-shot, One-shot, Few-shot

Output

ZERO-SHOT SUMMARY:

AI is transforming industries like healthcare, finance, and transportation by automating tasks and enabling data-driven decisions, such as early disease diagnosis and fraud detection.

While AI offers major benefits, it raises concerns about job losses and privacy, prompting governments to develop responsible, ethical AI policies.

ONE-SHOT SUMMARY:

Artificial Intelligence is reshaping healthcare, finance, education, and transportation through automation and data-driven insights, improving diagnosis, fraud detection, and safe navigation. Yet its growth raises concerns over job displacement and privacy, leading governments to craft ethical policies that balance innovation with accountability and shared benefits.

FEW-SHOT SUMMARY:

AI is revolutionizing global industries by automating tasks and enabling data-driven decisions in healthcare, finance, and transportation. Despite benefits like early diagnosis and fraud detection, concerns over job displacement and privacy persist, pushing governments to establish ethical guidelines that balance innovation with responsible, accountable AI use.

Comparison Table:

Technique	Accuracy	Completeness	Readability
Zero-shot	High	Moderate	High
One-shot	High	High	High
Few-shot	High	High	Very High

Observation: All three summaries stay close to the 50-word limit. The Zero-shot summary is accurate but slightly less polished. The One-shot summary improves completeness by following the format of the reference example. The Few-shot summary achieves the best balance of

accuracy, completeness, and readability, since it learns sentence structure and tone from two prior examples.

Comparison Table

Technique	Accuracy	Completeness	Readability
Zero-shot	High	Moderate	High
One-shot	High	High	High
Few-shot	High	High	Very High

Experiment No. 4

Professional Leave Request Email (Illness)

Aim

To design Zero-shot, One-shot, and Few-shot prompts to generate a professional email requesting leave due to illness, and to compare the generated emails based on tone, grammar, formatting, and completeness.

Algorithm

1. Start the program and import the required library for accessing the LLM API.
2. Initialize the API client with the model name and authentication key.
3. Define a function `generate_response()` that sends a prompt to the LLM and returns the generated text.
4. Construct a Zero-shot prompt that directly asks the model to write a leave application email due to illness.
5. Construct a One-shot prompt that provides one example leave/request email before asking for the new email.
6. Construct a Few-shot prompt that provides two example request emails before asking for the new email.
7. Send each prompt to `generate_response()` and store the three generated emails.
8. Display the Zero-shot, One-shot, and Few-shot email outputs.
9. Compare the three emails based on tone, grammar, formatting, and completeness, and tabulate the result.
10. Stop.

Python Program

```
import anthropic

client = anthropic.Anthropic(api_key="YOUR_API_KEY")

def generate_response(prompt, max_tokens=250):
    response = client.messages.create(
        model="claude-3-5-sonnet-20241022",
        max_tokens=max_tokens,
        messages=[{"role": "user", "content": prompt}]
    )
    return response.content[0].text.strip()

# ----- Zero-shot Prompt -----
zero_shot_prompt = """Write a professional email to my manager
requesting leave for 2 days due to illness."""

# ----- One-shot Prompt -----
```

```
one_shot_prompt = """Example:
Subject: Request for Work From Home
Dear Ma'am,
I hope this email finds you well. I would like to request permission
to work from home tomorrow due to a family commitment. I will ensure
all my tasks are completed on time and remain reachable throughout
the day.
Thank you for considering my request.
Regards,
Arjun
```

Now write a similar professional email requesting 2 days of leave due to illness."""

```
# ----- Few-shot Prompt -----
few_shot_prompt = """Example 1:
Subject: Request for Work From Home
Dear Ma'am,
I hope this email finds you well. I would like to request permission
to work from home tomorrow due to a family commitment. I will ensure
all my tasks are completed on time and remain reachable throughout
the day.
Thank you for considering my request.
Regards,
Arjun
```

```
Example 2:
Subject: Request for Early Leave
Dear Sir,
I am writing to request permission to leave the office two hours
early today due to a medical appointment. I will complete my pending
tasks before I leave and will be available on email if needed.
Thank you for your understanding.
Regards,
Sneha
```

Now write a similar professional email requesting 2 days of leave due to illness."""

```
print("ZERO-SHOT OUTPUT:\n", generate_response(zero_shot_prompt))
print("\nONE-SHOT OUTPUT:\n", generate_response(one_shot_prompt))
print("\nFEW-SHOT OUTPUT:\n", generate_response(few_shot_prompt))
```

Input

Recipient : Manager
Reason for Leave : Illness
Duration : 2 days

Prompting Techniques Used : Zero-shot, One-shot, Few-shot

Output

ZERO-SHOT OUTPUT:

Subject: Leave Request Due to Illness

Dear Sir/Madam,

I am writing to inform you that I am unwell and need to take leave for 2 days, starting tomorrow, to recover. I will make sure my pending work is handed over and will stay reachable over email in case of urgent matters.

Kindly approve my leave request. Thank you for your understanding.

Regards,
[J KARTHIK]

ONE-SHOT OUTPUT:

Subject: Request for Sick Leave

Dear Ma'am,

I hope this email finds you well. I am writing to request 2 days of leave starting tomorrow, as I am unwell and need time to recover. I will ensure that my ongoing tasks are handed over to a colleague and will remain reachable over email for anything urgent.

Thank you for considering my request.

Regards,
[J KARTHIK]

FEW-SHOT OUTPUT:

Subject: Request for Sick Leave

Dear Sir,

I am writing to request 2 days of leave, from [10-08-2026] to [12-08-2026], due to illness. I am currently unwell and need this time to rest and recover fully before returning to work. I will complete any urgent tasks before my leave begins and remain available over email for anything that cannot wait.

Thank you for your understanding and support.

Regards,
[J KARTHIK]

Comparison Table:

Technique	Tone	Grammar	Formatting	Completeness
Zero-shot	Polite	Correct	Basic	Moderate
One-shot	Warm/Formal	Correct	Structured	High
Few-shot	Formal/Warm	Correct	Well-structured	Very High

Observation: The Zero-shot email is grammatically correct but brief. The One-shot email adopts a warmer, more personal tone similar to the given example. The Few-shot email is the most complete, with clear dates, a proper handover statement, and consistent formatting learned from both reference examples.

Comparison Table

Technique	Tone	Grammar	Formatting
Zero-shot	Polite	Correct	Basic
One-shot	Warm / Formal	Correct	Structured
Few-shot	Formal / Warm	Correct	Well-structured

Experiment No. 5

Promotional Social Media Post for an AI Workshop

Aim

To design Zero-shot, One-shot, and Few-shot prompts to generate a promotional social media post for an AI Workshop using an LLM.

Algorithm

1. Start the program and import the required library for accessing the LLM API.
2. Initialize the API client with the model name and authentication key.
3. Define a function `generate_response()` that sends a prompt to the LLM and returns the generated text.
4. Construct a Zero-shot prompt that directly asks the model to write a promotional social media post for an AI workshop.
5. Construct a One-shot prompt that provides one example promotional post for a different event before asking for the new post.
6. Construct a Few-shot prompt that provides two example promotional posts before asking for the new post.
7. Send each prompt to `generate_response()` and store the three generated posts.
8. Display the Zero-shot, One-shot, and Few-shot outputs.
9. Compare the outputs for engagement, use of hashtags/emojis, and call-to-action strength.
10. Stop.

Python Program

```
import anthropic

client = anthropic.Anthropic(api_key="YOUR_API_KEY")

def generate_response(prompt, max_tokens=200):
    response = client.messages.create(
        model="claude-3-5-sonnet-20241022",
        max_tokens=max_tokens,
        messages=[{"role": "user", "content": prompt}]
    )
    return response.content[0].text.strip()

# ----- Zero-shot Prompt -----
zero_shot_prompt = """Write a promotional social media post for an
AI Workshop happening next month."""

# ----- One-shot Prompt -----
one_shot_prompt = """Example:
Event: Web Development Bootcamp
```

Post: "Ready to build your first website? Join our 3-day Web Development Bootcamp and learn HTML, CSS, and JavaScript from industry experts! Limited seats available. Register now!
#WebDev #Bootcamp #LearnToCode"

Now write a similar promotional social media post for an AI Workshop happening next month.""""

```
# ----- Few-shot Prompt -----  
few_shot_prompt = """Example 1:  
Event: Web Development Bootcamp  
Post: "Ready to build your first website? Join our 3-day Web  
Development Bootcamp and learn HTML, CSS, and JavaScript from  
industry experts! Limited seats available. Register now!  
#WebDev #Bootcamp #LearnToCode"
```

```
Example 2:  
Event: Data Science Meetup  
Post: "Curious about Data Science? Join our free meetup to explore  
real-world data projects, network with experts, and kickstart your  
data career! Seats are filling fast - reserve yours today!  
#DataScience #TechMeetup #CareerGrowth"
```

Now write a similar promotional social media post for an AI Workshop happening next month.""""

```
print("ZERO-SHOT OUTPUT:\n", generate_response(zero_shot_prompt))  
print("\nONE-SHOT OUTPUT:\n", generate_response(one_shot_prompt))  
print("\nFEW-SHOT OUTPUT:\n", generate_response(few_shot_prompt))
```

Input

Event Name : AI Workshop

Timing : Next month

Prompting Techniques Used : Zero-shot, One-shot, Few-shot

Output

ZERO-SHOT OUTPUT:

Excited to announce our upcoming AI Workshop next month! Dive into the world of Artificial Intelligence, learn from expert trainers, and get hands-on experience with real projects. Whether you're a beginner or a pro, this workshop has something for you. Don't miss out - seats are limited! #AIWorkshop #ArtificialIntelligence

ONE-SHOT OUTPUT:

"Ready to dive into the world of AI? Join our AI Workshop next month and learn Machine Learning, Deep Learning, and real-world AI applications from industry experts! Limited seats available.

Register now!

#AIWorkshop #ArtificialIntelligence #LearnAI"

FEW-SHOT OUTPUT:

"Curious about Artificial Intelligence? Join our hands-on AI Workshop next month to explore real-world AI projects, learn from industry experts, and kickstart your career in AI! Seats are filling fast - reserve yours today!

#AIWorkshop #ArtificialIntelligence #TechEvent #CareerGrowth"

Observation: The Zero-shot post is generic but informative. The One-shot post picks up the exact structure and hashtag style of the reference example. The Few-shot post is the most engaging, blending the tone, call-to-action urgency, and hashtag pattern learned from both reference examples.

